

IPSB / RDQM Health Check Framework Overview

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1. Executive Summary

- The IPSB/RDQM Health Check (HC2) framework provides structured, configuration-driven monitoring of IBM MQ RDQM environments.
- It runs independently on each node and writes standardized health fragments to shared storage.
- An aggregation process consolidates node fragments into instance-level health outputs for monitoring tools such as Zabbix.
- The system supports queue depth monitoring, DLQ alerting, SDR channel validation, and flexible routing by assignment group.
- Design philosophy emphasizes operational clarity, single-source-of-truth findings, and configuration-based behavior.

2. Architecture Overview

- Node Runner: Executes health checks locally on each RDQM node and produces fragment JSON files.
- Shared State Storage: Maintains queue_seen and queue_flow state files across failover events.
- Instance Aggregator: Consolidates node fragments into a single instance-level health file.
- Zabbix Integration: Consumes the instance-level output and routes alerts based on priority and assignment group.

3. Queue Monitoring Capabilities

- Local Queue Discovery via DIS QLOCAL(*)
- Queue Depth Threshold Monitoring.
- Configurable Stuck Queue Detection using consecutive-run streak logic.
- Dead Letter Queue (DLQ) Immediate Alerting (Depth ≥ 1 triggers P2).
- Pattern-Based Rule Engine for flexible per-QM and per-queue routing.
- Single Finding Per Queue (one priority, one assignment group).

4. Channel Monitoring (SDR)

- Discovery of defined SDR channels.
- Runtime status validation via DIS CHSTATUS.
- Configurable required channel validation.
- Separation of defined vs runtime status for operational clarity.

5. Rule Engine Layer (Implemented)

- Exact queue override support (highest precedence).
- Pattern-based rule support (e.g., *.DLQ).
- Per-QM and global defaults with clear precedence ordering.
- DLQ policy example: Depth ≥ 1 triggers P2 routed to the appropriate assignment group.
- Guarantees one queue generates exactly one finding.

6. Successfully Implemented to Date

- Node-level fragment generation.
- Shared-state continuity across RDQM failover.
- Queue discovery and depth evaluation.
- Streak-based stuck queue detection.
- DLQ immediate P2 alert model.
- Assignment group routing per queue.
- Config-driven behavior without hardcoding.
- Zabbix-ready structured JSON output.

7. Current Operational Model

- DLQs alert immediately when depth ≥ 1 .
- Non-DLQ queues default to configurable streak-based monitoring.

- All alert routing controlled via configuration.
- State files update each run; last_change only updates on depth changes.
- Framework designed for incremental enhancement without redesign.